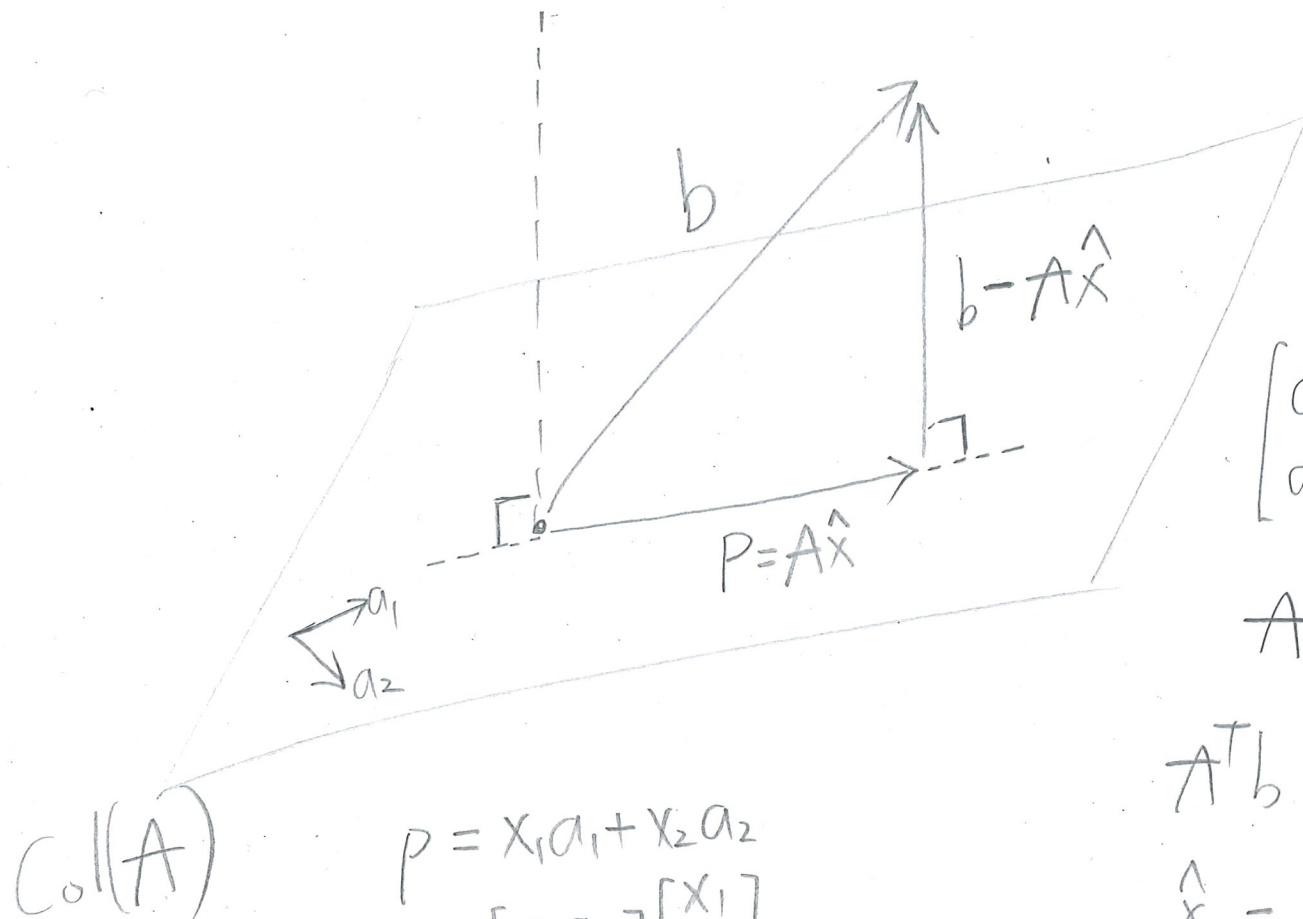




$$a_1^T (b - A \hat{x}) = 0$$

$$a_2^T (b - A \hat{x}) = 0$$

$$\begin{bmatrix} a_1^T \\ a_2^T \end{bmatrix} (b - A \hat{x}) = 0$$

$$A^T (b - A \hat{x}) = 0$$

$$A^T b = A^T A \hat{x}$$

$$\hat{x} = (A^T A)^{-1} A^T b$$

$$p = A \hat{x} = \underline{\underline{A (A^T A)^{-1} A^T b}}$$

$$P = A (A^T A)^{-1} A^T$$

$$\underline{\underline{p = P b}}$$